

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XIX: THE ODD CYCLIC CLASS THAT DOES NOT EXIST, AND WHAT THE EXCEPTIONAL ZERO SEES INSTEAD

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ABSTRACT. Paper XIV showed that the finite-dimensional commutative spectral triple of Paper II has $HC^1 = 0$, so no odd cyclic cocycle there can pair to the graph $\mathcal{L}$ -invariant $-\|h_\alpha\|^2$, and asked whether an infinite-dimensional non-commutative algebra would help. Three answers, in increasing order of finality. First, the algebra usually proposed, $\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$, is Morita-trivial: the $\mathbb{Z}$ -cover has vertex set $V \times \mathbb{Z}$ with the deck group acting freely on the second factor, so $\mathcal{A}_\alpha \cong \bigoplus_{v \in V} \mathcal{K}$, with $K_1 = 0$ and no odd K -theory to pair with at all. Second, the algebra where Bloch theory does live is the $\mathbb{Z}$ -equivariant one, $M_n(C(S^1))$, with $K_1 = \mathbb{Z}$; but its natural class, the voltage pencil $D(t) = (q+1)I - A(t)$, is *not invertible*, because $D(1)$ is the ordinary Laplacian. On a circle where the pencil is invertible the odd pairing is the winding number of $\det D$, and that number is the *order* of the exceptional zero — equal to 2 for every voltage class — whereas $\kappa_0 \mathcal{L}$ is its leading *coefficient*, $\frac{1}{2} \Theta''(0) = -\kappa_0 \|h_\alpha\|^2$. Index theory sees orders, not coefficients. Third, and this is a theorem rather than a diagnosis: an index pairing is additive in each slot, so if $\alpha \mapsto \tau_\alpha$ and $\alpha \mapsto [u_\alpha]$ are both linear and K_1 has rank one — as it does for $M_n(C(S^1))$ and for the shadow algebra alike — then $\langle \tau_\alpha, [u_\alpha] \rangle$ is a product of two linear forms in α . It cannot be $\kappa_0 \|h_\alpha\|^2$, which is positive definite on $H^1(Y; \mathbb{R})$, as soon as $b_1(Y) \geq 2$. So the impossibility of Paper XIV is not an artefact of finite dimensions. Five checks, on the K_4 and K_5 voltage classes.

1. THE PROPOSED ALGEBRA IS THE COMPACTS

Let Y be a finite connected $(q+1)$ -regular graph on n vertices, α an integer 1-cochain with $[\alpha] \neq 0$, and $\tilde{Y}_\alpha$ the associated $\mathbb{Z}$ -cover, with deck group $\mathbb{Z}$. Write κ_0 for the number of spanning trees of Y , h_α for the harmonic representative of $[\alpha]$, and $\mathcal{L}(Y_\bullet) = -\|h_\alpha\|^2$ [XIV, Def. 1.3].

Proposition 1.1. *The deck group acts freely on the vertex set of $\tilde{Y}_\alpha$ with exactly n orbits, and*

$$\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z} \cong \bigoplus_{v \in V} \mathcal{K}(\ell^2(\mathbb{Z})).$$

Consequently $K_0(\mathcal{A}_\alpha) \cong \mathbb{Z}^n$, $K_1(\mathcal{A}_\alpha) = 0$, and every odd cyclic cocycle pairs trivially.

Proof. A connected $\mathbb{Z}$ -cover of a connected graph has vertex set $V \times \mathbb{Z}$ with the deck group translating the second coordinate; the action is free and the orbits are the fibres, n of them (check (A)). Hence $C_0(\tilde{Y}_\alpha) \cong c_0(V) \otimes c_0(\mathbb{Z})$ equivariantly, with $\mathbb{Z}$ acting only on the second factor, and $c_0(V) \otimes (c_0(\mathbb{Z}) \rtimes \mathbb{Z}) \cong c_0(V) \otimes \mathcal{K}(\ell^2(\mathbb{Z}))$, the crossed product of c_0 of a free transitive action being the compacts. The K -theory and the vanishing of HC^{odd} for a direct sum of copies of $\mathcal{K}$ are standard. $\square$

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So the premise that $\mathcal{A}_\alpha$ “has nontrivial K_1 and HC^1 ” is false: it is stably $\mathbb{C}^n$, and its only cyclic cohomology is the even part, spanned by the n traces. The non-commutativity of a crossed product by a *free* action is illusory.

2. THE FLOQUET ALGEBRA, AND WHAT ITS PAIRING SEES

The algebra that does carry Bloch theory is the commutant, not the crossed product. After Floquet transform the $\mathbb{Z}$ -equivariant bounded operators on $\ell^2(\tilde{Y}_\alpha)$ are $M_n(C(S^1))$, whose K_1 is $\mathbb{Z}$, generated by the winding of a determinant. The natural element of it is the voltage pencil $D(t) = (q+1)I - A(t)$ of [IV, Lem. 1.1], whose Floquet determinant is Θ .

Proposition 2.1. *D is not invertible in $M_n(C(S^1))$: $\det D(1) = 0$, since $D(1)$ is the ordinary Laplacian of Y . The zero has order exactly two, and $\frac{1}{2}\Theta''(0) = -\kappa_0\|h_\alpha\|^2$ [IX, Thm. 1.2]. Hence the class $[D] \in K_1$ does not exist.*

Corollary 2.2 (What the pairing does see). *Let ε be small enough that the circle $|t-1| = \varepsilon$ meets no other zero of $\det D$. There the pencil is invertible and defines a class in K_1 , and its pairing with the generator of $HC^1(C^\infty(S^1))$ is the winding number of $\det D$. That number is the order of the exceptional zero, namely 2, for every voltage class; it does not depend on α , and so cannot be $\kappa_0\|h_\alpha\|^2$.*

Check (W) computes this winding number for the three K_4 classes and two K_5 classes and finds 2 in all five, against values of $\kappa_0\|h_\alpha\|^2$ equal to 8, 16, 24, 75 and 175 (Table 1). The contrast is the whole point: an index pairing is a topological degree, and a degree reads the *order* of vanishing; $\mathcal{L}$ is the *coefficient* at the same point.

	b_1	κ_0	$\ h_\alpha\ ^2$	$\kappa_0\ h_\alpha\ ^2$	winding of $\det D$
$K_4, \alpha = (1, 0, 0)$	3	16	1/2	8	2
$K_4, \alpha = (1, 1, 1)$	3	16	1	16	2
$K_4, \alpha = (1, 2, 0)$	3	16	3/2	24	2
K_5 , single edge	6	125	3/5	75	2
K_5 , triangle	6	125	7/5	175	2

TABLE 1. The cleared pencil has $[T^0] = [T^1] = 0$ and $[T^2] = -\kappa_0\|h_\alpha\|^2$ in every case (check (P)); the winding number is 2 in every case (check (W)).

Remark 2.3 (Where the integrality comes from). The task specification asked why the index should “absorb” $\kappa_0 = \det G$. It does not, and no index is involved: by [XIV, Thm. 7.1], $\|h_\alpha\|^2 = a_c^\top G^{-1} a_c$ with G the cycle Gram matrix and $a_c = C^\top \alpha$ integral, so

$$\kappa_0\|h_\alpha\|^2 = a_c^\top (\det G \cdot G^{-1}) a_c = a_c^\top \operatorname{adj}(G) a_c \in \mathbb{Z},$$

the adjugate of an integer matrix being an integer matrix. Integrality is Cramer’s rule, not an index theorem (check (I)).

3. THE OBSTRUCTION

Theorem 3.1 (No linear construction, in any algebra with K_1 of rank one). *Let A be a C^* -algebra with $\operatorname{rank}_{\mathbb{Z}} K_1(A) = 1$. Suppose given group homomorphisms*

$$H^1(Y; \mathbb{Z}) \rightarrow HC^1(A), \quad \alpha \mapsto \tau_\alpha, \quad H^1(Y; \mathbb{Z}) \rightarrow K_1(A), \quad \alpha \mapsto [u_\alpha].$$

If $b_1(Y) \geq 2$ then $\langle \tau_\alpha, [u_\alpha] \rangle \neq \kappa_0\|h_\alpha\|^2$ for some α .

Proof. Torsion classes pair trivially with a $\mathbb{C}$ -valued cocycle, so after tensoring with $\mathbb{Q}$ we may write $[u_\alpha] = \ell(\alpha)g$ for a fixed generator g of $K_1(A) \otimes \mathbb{Q} \cong \mathbb{Q}$ and a linear form ℓ on $H^1(Y; \mathbb{Q})$. Additivity of the pairing in the second slot gives $\langle \tau_\alpha, [u_\alpha] \rangle = \ell(\alpha) \mu(\alpha)$ with $\mu(\alpha) = \langle \tau_\alpha, g \rangle$, linear by additivity in the first slot. So the pairing is a product of two linear forms. If $b_1 \geq 2$ then $\ker \ell \neq 0$, so the product vanishes at some α with $[\alpha] \neq 0$; but $\kappa_0 \|h_\alpha\|^2 > 0$ there, since $h_\alpha = 0$ only for $[\alpha] = 0$. $\square$

Corollary 3.2. *The theorem applies to $M_n(C(S^1))$, where $K_1 \cong \mathbb{Z}$, and to the shadow algebra $\mathcal{O}_{B(Y)}$ of [VI], where $K_1 \cong \mathbb{Z}$ by [VI, Thm. 2.1]. Both candidate algebras of the task specification are therefore excluded, for every base graph with $b_1 \geq 2$ — in particular for K_4 ($b_1 = 3$) and K_5 ($b_1 = 6$), the graphs on which the $\mathcal{L}$ -invariant was computed.*

Check (Q) verifies the hypothesis that makes the argument bite: $\kappa_0 \|h_\alpha\|^2$ is the quadratic form $\text{adj}(G)$, which is positive definite of rank exactly $b_1 - 3$ for K_4 and 6 for K_5 — so it is not a product of two linear forms.

Remark 3.3 (What would be needed). Theorem 3.1 is not a proof that no odd cyclic class anywhere computes $\mathcal{L}$; it rules out the linear constructions in rank-one K_1 . Two escapes remain, and we state them as the honest residue of [XIV, Rem. 9.1(a)]. *Either* one finds an algebra whose K_1 has rank at least b_1 , with a natural $H^1(Y; \mathbb{Z}) \rightarrow K_1$ — the pairing may then be a genuine quadratic form; *or* one abandons linearity, in which case the assignment $\alpha \mapsto \tau_\alpha$ is not a cohomological construction but part of the input, and the exercise is circular.

The first escape is a real question and it has an obvious candidate, which we record because the rank matches exactly. The fundamental group of a connected graph is free of rank b_1 , and by Pimsner–Voiculescu

$$K_0(C_r^*(F_m)) \cong \mathbb{Z}, \quad K_1(C_r^*(F_m)) \cong \mathbb{Z}^m,$$

so $C_r^*(\pi_1(Y)) = C_r^*(F_{b_1})$ has rank $K_1 = b_1$ — precisely the threshold at which Theorem 3.1 stops applying, and the smallest rank at which a positive definite form on $H^1(Y; \mathbb{R})$ could be a pairing at all. Whether the natural map $H^1(Y; \mathbb{Z}) \cong \mathbb{Z}^{b_1} \rightarrow K_1(C_r^*(F_{b_1})) \cong \mathbb{Z}^{b_1}$ and a family τ_α can be chosen so that the pairing is $\kappa_0 \|h_\alpha\|^2$ we do not know; we note only that $\kappa_0 \|h_\alpha\|^2 = a_c^\top \text{adj}(G) a_c$ is already a quadratic form on that lattice, so the shape is right and only the construction is missing.

4. VERIFICATION

key	check	mode
(A)	Proposition 1.1: free deck action, n orbits	exact
(P)	the cleared pencil: $[T^0] = [T^1] = 0$, $[T^2] = -\kappa_0 \ h_\alpha\ ^2$	exact
(W)	Corollary 2.2: the winding number is 2 for every α	numeric
(I)	Remark 2.3: $\kappa_0 \ h_\alpha\ ^2 = a_c^\top \text{adj}(G) a_c \in \mathbb{Z}$	exact
(Q)	$\text{adj}(G)$ is positive definite of rank $b_1 \geq 2$	exact

TABLE 2. The verification battery (`cyclic_obstruction.py`), 5/5, about one second.

5. STATUS, SCOPE, CORRECTIONS

Proved. Proposition 1.1, Proposition 2.1 (given [IX, Thm. 1.2]), Corollary 2.2, Theorem 3.1 and Corollary 3.2.

Not proved. That no odd cyclic class in any algebra computes $\mathcal{L}$. What is proved is that the two algebras named, and every algebra with K_1 of rank one, fail for a structural reason, and that

the natural class in the Floquet algebra fails for a different and more interesting one — it does not exist, and the object obstructing it is the exceptional zero whose coefficient is the invariant sought.

Remark 5.1 (Solved, open, impossible). *Solved*: the two candidate algebras of [XIV, Rem. 9.1(a)] are ruled out, the first because it is Morita-trivial and the second by Theorem 3.1. *Open*: an algebra with K_1 of rank $\geq b_1$ and a natural map from $H^1(Y; \mathbb{Z})$ (Remark 3.3); and, unchanged from [XIV], the question whether $\mathcal{L}$ has any index-theoretic description at all, as opposed to the spectral-action and Albanese descriptions it is already known to have [XIV, Cor. 7.2]. *Impossible*: an odd pairing linear in α against a rank-one K_1 , whenever $b_1 \geq 2$. Together with [XIV, Prop. 2.1] this closes the naive routes in both the finite-dimensional and the infinite-dimensional directions.

Remark 5.2 (Corrections to the task specification). Three. (1) “ $\mathcal{A}_\alpha = C_0(\tilde{Y}_\alpha) \rtimes \mathbb{Z}$ has nontrivial K_1 and HC^1 ” is false; the action is free, so the crossed product is a sum of copies of the compacts (Proposition 1.1). The specification’s own hint is the obstruction to following it. (2) The proposed mechanism — pair a cocycle built from the integral period vector a_c with the unit shift $u \in \mathbb{Z}$ and read off $\|h_\alpha\|^2$ from a heat-kernel limit — is linear in α in the K_1 slot and cannot produce a positive definite quadratic form (Theorem 3.1); this is the same objection [XIV, Prop. 2.1(iii)] raised against the finite-dimensional proposal, and it survives the passage to infinite dimensions unchanged. (3) There is nothing for an index to “absorb”: $\kappa_0 \|h_\alpha\|^2 = a_c^\top \text{adj}(G) a_c$ is an integer by Cramer’s rule (Remark 2.3), and no index theorem is needed to see it.

PROVENANCE

Unrefereed preprint. Every numbered statement is proved above. The computations verify the exceptional-zero coefficient and the winding number independently on five voltage classes, and confirm the positive definiteness that makes Theorem 3.1 bite. The winding number was computed before the theorem was formulated, in order to find out what the available pairing actually measures.

REFERENCES

- [1] R. Chen, *Algorithmic non-commutative class field theory, II: the trace obstruction and the K-homological bulk*, 2026.
- [2] R. Chen, *Algorithmic non-commutative class field theory, IV: the limit module of abelian towers and the characteristic identity*, 2026.
- [3] R. Chen, *Algorithmic non-commutative class field theory, VI: the shadow algebra*, 2026.
- [4] R. Chen, *Algorithmic non-commutative class field theory, IX: exceptional zeros, the graph $\mathcal{L}$ -invariant, and the operator-algebraic Mazur–Tate–Teitelbaum formula*, 2026.
- [5] R. Chen, *Algorithmic non-commutative class field theory, XIV: the graph $\mathcal{L}$ -invariant as spectral action, Bloch effective mass and Albanese length, and what the cyclic Chern character sees*, 2026.
- [6] A. Connes, *Noncommutative Geometry*, Academic Press, 1994.
- [7] M. Pimsner and D. Voiculescu, *K-groups of reduced crossed product C^* -algebras by free groups*, J. Operator Theory **8** (1982), 131–156.

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